

MAT110

Differential Calculus and Coordinate Geometry

Lecture 11

Chain Rule for Partial Derivatives and Extrema

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1 Chain Rule for Partial Derivatives

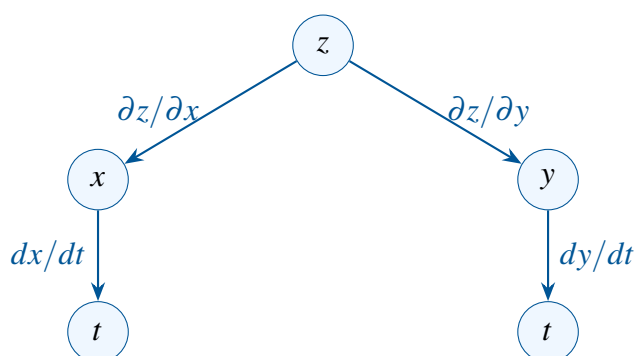
Reference: H. Anton, Calculus, Section 13.5.

Chain Rule: Two Dependent Variables

If $x = x(t)$ and $y = y(t)$ are differentiable at t , and if $z = f(x, y)$ is differentiable at the point $(x, y) = (x(t), y(t))$, then $z = f(x(t), y(t))$ is differentiable at t and

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}.$$

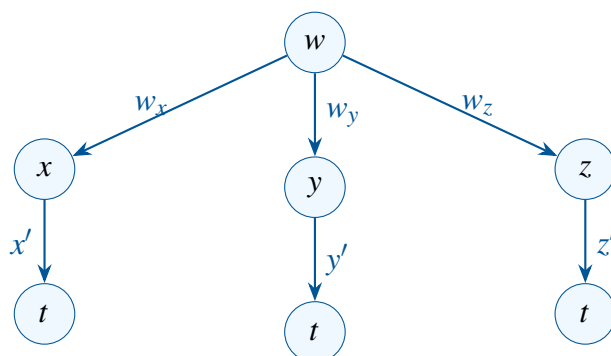
The ordinary derivatives are evaluated at t , and the partial derivatives are evaluated at (x, y) .



Chain Rule: Three Dependent Variables

Let $w = f(x, y, z)$, where $x = x(t)$, $y = y(t)$, and $z = z(t)$ are differentiable at t . Then

$$\frac{dw}{dt} = \frac{\partial w}{\partial x} \frac{dx}{dt} + \frac{\partial w}{\partial y} \frac{dy}{dt} + \frac{\partial w}{\partial z} \frac{dz}{dt}.$$



Example 01

If

$$z = 3x^2y^3, \quad x = t^4, \quad y = t^2,$$

evaluate dz/dt using chain rule.

Solution

By the chain rule,

$$\begin{aligned}\frac{dz}{dt} &= \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt} \\ &= (6xy^3)(4t^3) + (9x^2y^2)(2t) \\ &= 6(t^4)(t^2)^3(4t^3) + 9(t^4)^2(t^2)^2(2t) \\ &= 24t^{13} + 18t^{13} = 42t^{13}.\end{aligned}$$

Thus,

$$\boxed{\frac{dz}{dt} = 42t^{13}}.$$

Example 02

If

$$z = 3 \cos x - \sin(xy), \quad x = \frac{1}{t}, \quad y = 3t,$$

evaluate dz/dt using chain rule.

Solution

Here z is a function of the two independent variables x and y , while x and y are functions of the one independent variable t . Hence,

$$\begin{aligned}\frac{dz}{dt} &= \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt} \\ &= (-3 \sin x - y \cos xy) \left(-\frac{1}{t^2}\right) + (-x \cos xy)(3) \\ &= \frac{3 \sin(1/t) - 3t \cos(3) + 3t \cos(3)}{t^2}.\end{aligned}$$

Therefore,

$$\boxed{\frac{dz}{dt} = \frac{3 \sin(1/t)}{t^2}}.$$

Problem 03

(i) If $z = xye^y$, $x = e^{t^2}$, and $y = 5t$, evaluate dz/dt .

(ii) If $z = x^2y + xy^2$, $x = 3t$, and $y = t^2$, evaluate dz/dt .

Practice

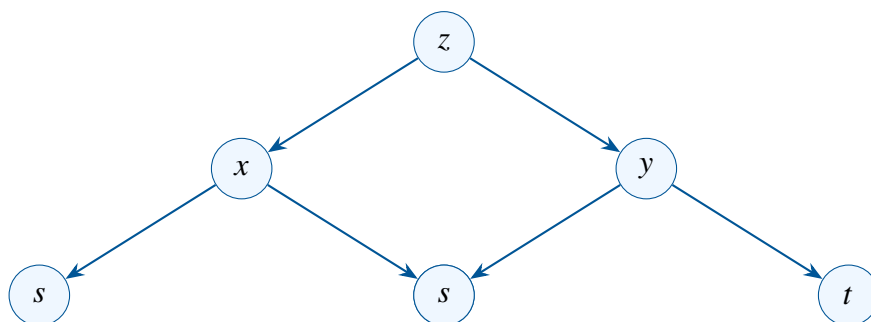
H. Anton, Section 13.5: Problem 8 (1–10).

1.1 Chain Rule with Two Independent Variables

Suppose that $z = f(x, y)$ is differentiable, where $x = x(s, t)$ and $y = y(s, t)$ are differentiable functions of s and t . Then

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial t},$$

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial s}.$$



Example 03

If

$$z = e^x \sin y, \quad x = st^2, \quad y = s^2 t,$$

find $\partial z / \partial s$ and $\partial z / \partial t$ using chain rule.

Solution

Using the dependency paths from z to s ,

$$\begin{aligned} \frac{\partial z}{\partial s} &= \frac{\partial z}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial s} \\ &= e^x \sin y \cdot t^2 + e^x \cos y \cdot 2st \\ &= t^2 e^{st^2} \sin(s^2 t) + 2st e^{st^2} \cos(s^2 t). \end{aligned}$$

Similarly,

$$\begin{aligned} \frac{\partial z}{\partial t} &= \frac{\partial z}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial t} \\ &= e^x \sin y \cdot 2st + e^x \cos y \cdot s^2 \\ &= 2st e^{st^2} \sin(s^2 t) + s^2 e^{st^2} \cos(s^2 t). \end{aligned}$$

[N.B. To find $\partial z / \partial s$ or $\partial z / \partial t$, form the product of the partial derivatives along each path from z to the required variable, and then add those products.]

2 Maxima and Minima of Functions of Two Variables

Reference: H. Anton, *Calculus*, Section 13.8.

Relative and Absolute Maximum (Definition 13.8.1)

A function f of two variables is said to have a *relative maximum* at (x_0, y_0) if there is a disk centered at (x_0, y_0) such that

$$f(x_0, y_0) \geq f(x, y)$$

for all points (x, y) that lie inside the disk.

The function f is said to have an *absolute maximum* at (x_0, y_0) if

$$f(x_0, y_0) \geq f(x, y)$$

for all points (x, y) in the domain of f .

Relative and Absolute Minimum (Definition 13.8.2)

A function f of two variables is said to have a *relative minimum* at (x_0, y_0) if there is a disk D centered at (x_0, y_0) such that

$$f(x_0, y_0) \leq f(x, y), \quad (x, y) \in D.$$

The function has an *absolute minimum* at (x_0, y_0) if

$$f(x_0, y_0) \leq f(x, y)$$

for every (x, y) in the domain of f .

Necessary Condition for a Relative Extremum (Theorem 13.8.4)

If f has a relative extremum at a point (x_0, y_0) and the first-order partial derivatives of f exist at this point, then

$$f_x(x_0, y_0) = 0 \quad \text{and} \quad f_y(x_0, y_0) = 0.$$

Critical Point (Definition 13.8.5)

A point (x_0, y_0) in the domain of a function $f(x, y)$ is called a critical point of the function if

$$f_x(x_0, y_0) = 0 \quad \text{and} \quad f_y(x_0, y_0) = 0,$$

or if one or both partial derivatives do not exist at (x_0, y_0) .

Saddle Point

A saddle point of a function is a point in the domain of the function where it neither attains a maximum value nor attains a minimum value.

Example 04: Critical Points

Find all the critical points of

$$f(x, y) = \frac{x^2}{2} - x + xy^2.$$

Solution

We have

$$f_x = x - 1 + y^2, \quad f_y = 2xy.$$

At a critical point,

$$x - 1 + y^2 = 0, \quad 2xy = 0.$$

The second equation gives either $x = 0$ or $y = 0$.

If $x = 0$, then $-1 + y^2 = 0$, so $y = \pm 1$, giving $(0, 1)$ and $(0, -1)$.

If $y = 0$, then $x - 1 = 0$, so $x = 1$, giving $(1, 0)$.

Therefore, the critical points are

$$(0, 1), (0, -1), (1, 0).$$

Second-Partial Test (Theorem 13.8.6)

Let f be a function of two variables with continuous second-order partial derivatives in some disk centered at a critical point (x_0, y_0) , and let

$$D = f_{xx}(x_0, y_0)f_{yy}(x_0, y_0) - (f_{xy}(x_0, y_0))^2.$$

Then:

- (a) If $D > 0$ and $f_{xx}(x_0, y_0) > 0$, then f has a relative minimum at (x_0, y_0) .
- (b) If $D > 0$ and $f_{xx}(x_0, y_0) < 0$, then f has a relative maximum at (x_0, y_0) .
- (c) If $D < 0$, then f has a saddle point at (x_0, y_0) .
- (d) If $D = 0$, then no conclusion can be drawn.

Example 05: Relative Extrema and Saddle Points

Locate all relative extrema and saddle points of

$$f(x, y) = 4xy - x^4 - y^4.$$

Solution

The first-order partial derivatives are

$$f_x = 4y - 4x^3, \quad f_y = 4x - 4y^3.$$

Setting both equal to zero gives

$$y = x^3, \quad x = y^3.$$

Therefore,

$$\begin{aligned} x &= y^3 = (x^3)^3 = x^9, \\ x^9 - x &= 0, \\ x(x^8 - 1) &= 0. \end{aligned}$$

Hence $x = 0$ or $x^8 = 1$, so $x = 0, 1, -1$. Since $y = x^3$, the critical points are

$$(0, 0), \quad (1, 1), \quad (-1, -1).$$

The second-order partial derivatives are

$$f_{xx} = -12x^2, \quad f_{yy} = -12y^2, \quad f_{xy} = 4.$$

Thus,

$$D = f_{xx}f_{yy} - f_{xy}^2 = 144x^2y^2 - 16.$$

No.	Critical point	D	f_{xx}	Conclusion
1	$(0,0)$	$-16 < 0$	—	Saddle point at $(0,0)$
2	$(1,1)$	$128 > 0$	$-12 < 0$	Relative maximum
3	$(-1,-1)$	$128 > 0$	$-12 < 0$	Relative maximum

Therefore, $(0,0)$ is a saddle point, and there are relative maxima at $(1,1)$ and $(-1,-1)$. Now, we need to evaluate $f(1,1)$ and $f(-1,-1)$.